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Fig. 3: Block diagram of the transmitter side



Fig. 4: Block diagram of the receiver side

mode. LCD pins RS, EN, D4, D5, D6 and D7 are connected to digital I/O pins 12, 11, 7, 8, 9 and 10 of Arduino, respectively.

434 MHz TX-RX modules. The 434MHz transmitter (TX) and receiver (RX) modules are used to transmit and receive radio frequency (RF) signals between two devices. Carrier frequency of the module used here is 434MHz.

The circuit diagrams of the transmitter and receiver sides are shown in

Figs 5 and 6, respectively.

The signal whose frequency is to be measured is applied to test signal input terminals, as shown in the transmitter-side circuit diagram (Fig. 5). In this frequency meter, test signal must be an alternating one and its amplitude must not be greater than 10V peak-to-peak.

The next stage is a bridge rectifier, which converts the alternating signal to a fully-rectified pulsating DC signal. The signal is then applied

List of Components

Table 1

Name	Quantity
LM7805, 5V regulator	1
MCT2E optocoupler IC	1
434MHz TX module	1
434MHz RX module	1
Arduino Uno board	1
Diode 1N4007	4
Resistor, 1k Ω , 1/4W	1
Resistor, 10k Ω , 1/4W	1
16 $\times$ 2 LCD	1
Preset, 10k Ω	1
Capacitors 10 μ F, 25V	2
9V battery	1

(All these are available from kitsnspares.com)

to pin 1 of MCT2E opto-coupler. Pin 1 is connected to anode of the internal LED of the opto-coupler. With the help of the opto-coupler, spikes are produced. Frequency of the signal is twice that of test signal input. The signal is applied to DATA pin of 434 MHz TX.